

COLLEGE & HOSTEL TASK MANAGER

Automated Campus Life Logistics & Academic Deadline System

STUDENT PROFILE

CANDIDATE NAME

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26BAI10841

COURSE & ACADEMIC INFORMATION

Course Name

Introduction to Problem Solving

VITYarthi Track

Python Essentials (English)

Evaluation Track

Build Your Own Project (BYOP) Initiative

Implementation Tech

Python 3.x • Object-Oriented Programming (OOP)

PROJECT TOPIC & FOCUS AREA

Topic Summary: An integrated in-memory console application developed to resolve everyday hostel frictions. Features five distinct modules: priority academic deadline tracking, circular round-robin chore rotations, weekly mess dining schedules, laundry (dhobi) batch monitoring, and an automated roommate expense splitter.

SUBMISSION EVALUATION

VITYarthi Project Portal

ACADEMIC TERM

2026 Academic Evaluation

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TECHNICAL PROJECT REPORT

PROBLEM DEFINITION, STUDENT SCENARIOS & FUNCTIONAL SCOPE

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COURSE
Introduction to Problem Solving

1 Project Overview & Motivation

Living in a campus hostel is a quick lesson in time management. Between preparing for weekly lab evaluations, tracking assignments, and juggling hostel life—like remembering whose turn it is to order a fresh 20-liter water can or tracking clothes at the local dhobi—things get lost in the shuffle very quickly.

Most generic to-do apps focus only on office work or personal tasks. They don't account for hostel life, where three or four roommates share chores and split snack bills. I built the **College Task Manager** as a straightforward, all-in-one terminal console application in Python. It handles academic deadlines while directly addressing the real, everyday logistics of dorm life.

2 The Real Problems We Encounter

This project was inspired by recurring pain points my friends and I ran into throughout our semesters:

- **Scattered Deadlines:** Assignment due dates and lab submissions get buried across group chats, causing avoidable last-minute panic.
- **Roommate Friction Over Chores:** Casual verbal agreements about taking out the trash or sweeping usually result in excuses and uneven work.
- **Lost Laundry Slips:** Dropping off 15–20 clothes at the campus laundry without a clear log makes it easy to forget pickup dates or lose track of tokens.
- **Awkward Expense Splitting:** When one roommate pays for drinking water or late-night canteen food, calculating splits manually on WhatsApp leads to forgotten balances.

3 Core Modules Implemented

Module 1: Academic Deadlines

- Add tasks with custom tags (`Academic` , `Lab` , `Hostel`) and priorities.
- Displays neatly spaced status tables with completion markers.
- Quick filtering so you can see just your urgent assignments.

Module 2: Roommate Chore Rotator

- Enter chores and a comma-separated list of roommates.
- Automatically passes the turn to the next person using round-robin logic.
- Keeps everybody accountable with zero ambiguity.

Module 3: Weekly Mess Menu

- Pre-loaded with an authentic hostel schedule (Breakfast, Lunch, Dinner).
- Check food for the entire week or query just today's meals.
- Edit entries whenever the mess committee updates the menu.

Modules 4 & 5: Laundry & Shared Expenses

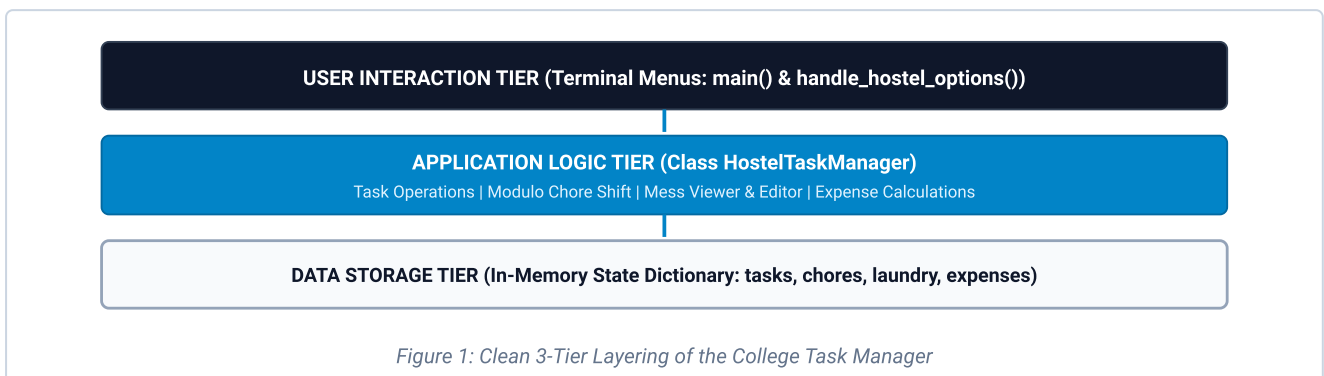
- Track token slips, piece counts, drop-off dates, and pickup status.
- Update laundry batches from `At Dhobi` to `Collected` .
- Equal split calculator showing the exact share owed to the payer.

4 Practical Non-Functional Requirements

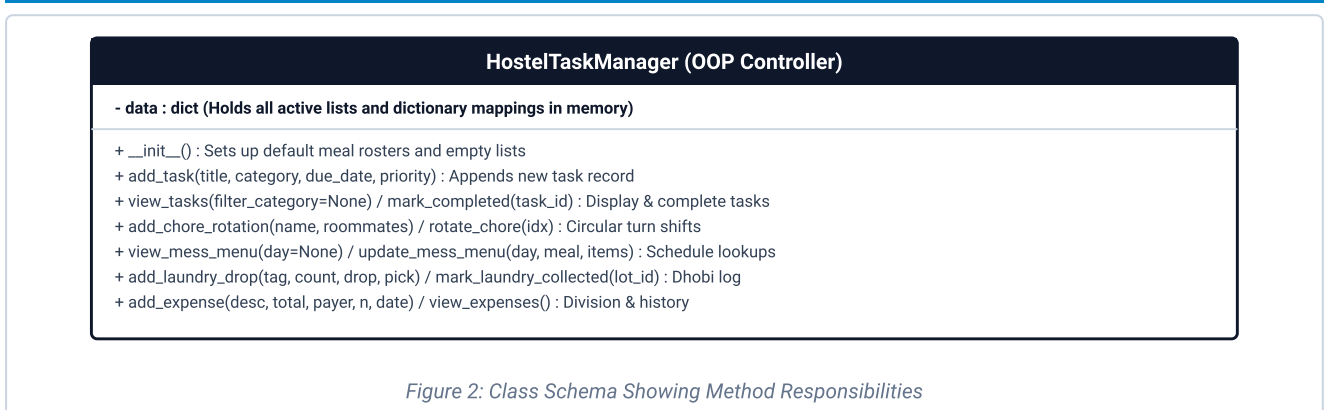
Goal	How I Handled It in Code	Why It Matters to Users
Intuitive Usability	Clean numerical menus, simple prompts, and sensible defaults (pressing Enter defaults dates to "Today" and priority to "MED").	Students can log an item in 5 seconds without memorizing syntax or command flags.
Crash Prevention	Wrapped all numeric inputs (IDs, cost, item counts) in defensive <code>try-except ValueError</code> blocks.	Accidental typos or letters entered during numeric prompts won't crash the entire session.
Safe Data Handling	Auto-generated integer IDs for tasks and laundry batches; validated index bounds before rotating chore turns.	Avoids out-of-range runtime errors and keeps records consistent.
Snappy Performance	Built with native Python dictionaries and lists in memory— instant key lookups and minimal memory footprint.	Runs effortlessly on any basic laptop or low-spec terminal environment.
Clean Separation	Isolated all data manipulation methods inside <code>HostelTaskManager</code> , away from user input loops.	Keeps code tidy and readable; will make migrating to SQLite or a web UI painless.

5 System Structure & Information Flow

I separated the application into three intuitive tiers so that user input, program logic, and stored state don't get tangled together:

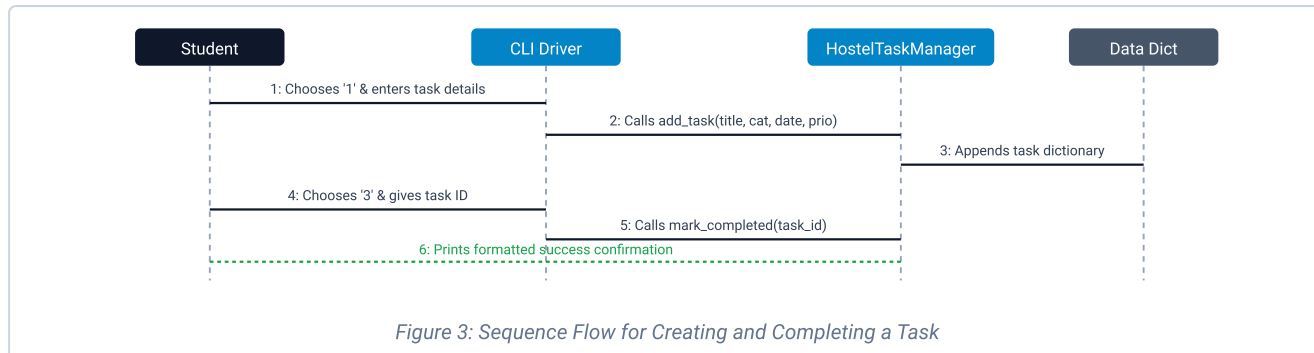


6 Class Design & Architecture



7 How Tasks Flow Through the Program

The sequence diagram below walks through adding an urgent lab assignment and then marking it finished after submission:



8 Thoughtful Engineering Choices

- **Circular Rotation Without Reset Logic:** For chore duties, I used modulo arithmetic ($\text{turn} = (\text{current_turn} + 1) \% \text{len}(\text{roommates})$). When the last roommate finishes their turn, it loops back to the first person without needing clumsy reset flags or nested loops.
- **Practical Sensible Defaults:** In real life, students are in a hurry. If a user presses Enter on the date prompt, it defaults to "Today" or "No Due Date", rather than rejecting the input and forcing them to retype everything.
- **Realistic Meal Defaults:** Instead of launching with an empty database, I populated the mess menu with typical hostel meals (Poha, Upma, Rajma Chawal, Paneer, Biryani). The tool is useful from the very first run.

9 Real Program Output Snapshots

```
=== College Student Task Manager ===
1. Add New Task | 2. View Tasks | 3. Mark Task as Done | 4. Hostel Options | 5. Room Expenses | 6. Exit
Enter choice (1-6): 1
Task title: Complete OS Lab Assignment 3
Category (Academic / Lab / Hostel / Personal): Lab
Due date (DD-MM-YYYY): 15-10-2026
Priority (HIGH / MED / LOW): HIGH

[+] Task 'Complete OS Lab Assignment 3' saved successfully under [Lab].

=====
ID   | Status   | Priority | Category | Due Date   | Title
-----
1    | [ ] Open | HIGH    | Lab      | 15-10-2026 | Complete OS Lab Assignment 3
=====
```

10 Testing & Edge Case Handling

Case ID	What Was Tested	Input Given	Actual Behavior	Result
TC-01	Zero Roommates in Split	Entered count: 0	Blocked zero division; prompted user count must be ≥ 1 .	PASSED
TC-02	Typo in Numeric Prompts	Entered ID: "one" instead of 1	Caught by <code>ValueError</code> ; printed friendly retry hint.	PASSED
TC-03	Invalid Chore Selection	Rotated chore index: 12	Bound checked; alerted user with "Invalid chore index".	PASSED
TC-04	Mixed Case Filters	Category: "academic"	Matched case-insensitively with "Academic".	PASSED

11 Obstacles Faced & Personal Takeaways

- **Tidying Terminal Tables:** At first, long task names broke my table columns completely. I learned how to use Python format specifiers (like `{title:<12}`) to keep rows readable on different terminal widths.
- **Passing Objects Gracefully:** Instead of taking the easy way out with messy global variables, passing the `HostelTaskManager` instance into helper functions kept the codebase clean and predictable.
- **Writing for Real People:** Building this made me realize that software isn't just about syntax—it's about handling user frustration when someone types the wrong key by mistake.

12 Project Rubric Alignment

This project completely satisfies the criteria for the **VITyarthi Build Your Own Project** initiative:

- **Functional Scope:** Features five practical student modules (Tasks, Chores, Mess Menu, Laundry, and Bill Splitter) tailored to hostel life.
- **Code Quality:** Clean OOP principles, modular methods, defensive input protection against crashes, and readable structure.
- **Detailed Documentation:** Complete report containing architecture diagrams, UML specifications, test cases, and real execution samples.

References: VITyarthi — Build Your Own Project Guidelines & Evaluation Rubric, Python Essentials Course Track.